Unit - III

Time & Work

Important facts and Formulae :-

① If A can do a piece of work in n days, then A's 1 day's work = $1/n$.

② If A's 1 day's work = $1/n$, then A can finish the work in n days.

③ If A is thrice as good a workman B, then:

Ratio of work done by A and B = 3 : 1

Ratio of times taken by A and B to finish a work = 1 : 3.

Ex:-1 Worker A takes 8 hours to do a job. Worker B takes 10 hours to do the same job. How long should it take both A and B, working together but independently, to do the same job?

Sol^n
A's 1 hour work = $1/8$
B's 1 hour work = $1/10$.

(A+B)'s 1 hour's work = $\left(\frac{1}{8} + \frac{1}{10}\right) = \frac{9}{40}$

∴ Both A and B will finish the work in $40/9 = 4\frac{4}{9}$ days.

Ex:-2 A and B together can complete a piece of work in 4 days. If A alone can complete the same work in 12 days, in how many days can B alone complete that work?

Ans (A+B)'s one day work = $1/4$

A's one day work = $1/12$.

$\therefore$ B's one day work $= \left(\frac{1}{4} - \frac{1}{12}\right) = \frac{1}{6}$.

Hence B alone can complete the work in 6 days.

Ex:-3 A can do a piece of work in 7 days of 9 hours each and B can do it in 6 days of 7 hours each. How long will they take to do it, working together $8\frac{2}{5}$ hours a day?

Soln A can complete the work in $(7\times9) = 63$ hours
B can complete the work in $(6\times7) = 42$ hours

$\therefore$ A's 1 hour work $= 1/63$

and B's 1 hour work $= 1/42$

(A+B)'s 1 hour work $= \frac{1}{63} + \frac{1}{42} = \frac{5}{126}$

Both will finish the work in $\frac{126}{5}$ hours

Number of days of $8\frac{2}{5}$ hrs each $= \frac{126}{5} \times \frac{5}{42} = 3$ days

Ex:-4 A and B can do a piece of work in 18 days, B and C can do it in 24 days; A and C can do it in 36 days. In how many days will A, B and C finish it, working together and seperately?

Sol^n $(A+B)$'s 1 day work $= \frac{1}{18}$,

$(B+C)$'s 1 day work $= 1/24$,
and $(A+C)$'s 1 day's work $= 1/36$.

On adding, we get.

$2(A+B+C)$'s 1 day work $= \frac{1}{18} + \frac{1}{24} + \frac{1}{36} = \frac{9}{72} = \frac{1}{8}$.

$\therefore$ $(A+B+C)$'s 1 day's work $= \frac{1}{16}$

Thus, A, B and together can finish the work in 16 days.

Now, A's one day work $= (A+B+C)$'s 1 day work $-$

$(B+C)$'s 1 day's work

$= \left(\frac{1}{16} - \frac{1}{24}\right) = \frac{1}{48}$.

A alone can finish the work in 48 ~~work~~. days.

Similarly, B's 1 day's work = $\left(\frac{1}{16} - \frac{1}{36}\right) = \frac{5}{144}$.

$\therefore$ B alone can finish the work in $\frac{144}{5} = 28\frac{4}{5}$ days.

And C's 1 day's work $= \left(\frac{1}{16} - \frac{1}{18}\right) = \frac{1}{144}$

$\therefore$ C alone can finish the work in 144 days.

Ex:-5 A is twice as good a workman as B and together they finish a piece of work in 18 days. In how many days will A alone finish the work?

Soln (A's 1 day's work) : (B's 1 day's work) = 2 : 1

(A + B)'s 1 day's work = 1/18.

Divide 1/18 in the ratio 2 : 1.

$\therefore$ A's 1 day's work $= \left(\frac{1}{18} \times \frac{2}{3}\right) = \frac{1}{27}$.

Hence, A alone can finish the work in 27 days.

Ex:-6 A can do a certain job in 12 days. B is 60% more efficient than A. How many days does B alone take to do the same job?

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Sol^n Ratio of times taken by A and B = 160 : 100

$= 8:5$.

Suppose B alone takes x days to do the job

Then, $8:5::12:x$

$\Rightarrow 8x = 5\times 12 \Rightarrow x = 7\frac{1}{2}$ days.

Ex:-7 A can do a piece of work in 80 days. He works at it for 10 days and then B alone finishes the remaining work in 42 days. In how much time will A and B, working together, finish the work?

Sol^n Work done by A in 10 days $= \left(\frac{1}{80}\times 10\right) = \frac{1}{8}$.

Remaining work $= \left(1-\frac{1}{8}\right) = \frac{7}{8}$.

Now, $7/8$ work is done by B in 42 days.

Whole work will be done by B in $\left(42\times\frac{8}{7}\right)$

$= 48$ days.

A's one day work $= 1/80$

B's one day work $= 1/48$.

$\therefore$ (A+B)'s 1 day's work $= \frac{1}{80}+\frac{1}{48} = \frac{8}{240} = \frac{1}{30}$.

Hence both will finish the work in 30 days.

Ex:-8 A and B undertake to do a piece of work for Rs. 600. A alone can do it in 6 days while B alone can do it in 8 days. With the help of C, they finish it in 3 days. Find the share of each.

Soln C's 1 day's work $= \frac{1}{3} - \left(\frac{1}{6} + \frac{1}{8}\right) = \frac{1}{24}$.

$\therefore$ A : B : C = Ratio of their 1 day's work

$\frac{1}{6} : \frac{1}{8} : \frac{1}{24} = 4:3:1$.

$\therefore$ A's share = Rs. $\left(600 \times \frac{4}{8}\right)$ = Rs. 300

B's share = Rs $\left(600 \times \frac{3}{8}\right)$ = Rs. 225.

C's share = Rs. $\left(600 - (300 - 225)\right)$ = Rs. 75.

Ex:-9 A and B working separately can do a piece of work in 9 and 12 days respectively. If they work for a day alternately A ~~bee~~ beginning, in how many days, the work will be completed?

Soln (A + B)'s 2 day's work $= \left(\frac{1}{9} + \frac{1}{12}\right) = \frac{7}{36}$.

Work done in 5 pairs of days $= \left(5 \times \frac{7}{36}\right) = \frac{35}{36}$.

Remaining work $= \left(1 - \frac{35}{36}\right) = \frac{1}{36}$.

On 11^{th} day, it is A's turn, $\frac{1}{9}$ work is done by him 1 day.

$\frac{1}{36}$ work is done by him in $9 \times \frac{1}{36} = \frac{1}{4}$ days.

$\therefore$ Total time taken $= \left(10 + \frac{1}{4}\right)$ days

$= 10\frac{1}{4}$ days.

Ex:-10 45 men can complete a work in 16 days. Six days after they started working, 30 more men joined them. How many days will they now take to complete the remaining work?

Sol^n (45×16) men can complete the work in 1 day.

$\therefore$ 1 men's 1 day's work $= \frac{1}{720}$

45 men's 6 day's work $= \left(\frac{1}{16} \times 6\right) = \frac{3}{8}$,

Remaining work $= \left(1 - \frac{3}{8}\right) = \frac{5}{8}$.

75 men's 1 day's work $= \frac{75}{720} = \frac{5}{48}$.

Now, 5/48 work is done by them in 1 day.

$\therefore \frac{5}{8}$ work is done by them in $\left(\frac{48}{5} \times \frac{5}{8}\right) = 6$ days.

Ex 1-11 2 men and 3 boys can do a piece of work in 10 days while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy do the work?

Soln Let 1 men's 1 day's work = x

and 1 boy's 1 day's work = y.

Then, $2x + 3y = \frac{1}{10}$ and $3x + 2y = \frac{1}{8}$.

Solving, we get : $x = \frac{7}{200}$ and $y = \frac{1}{100}$.

∴ (2 men + 1 boy)'s 1 day's work $= \left(2 \times \frac{7}{200} + 1 \times \frac{1}{100}\right)$

$= \frac{16}{200} = \frac{2}{25}$.

So, 2 men and 1 boy together can finish the work in $\frac{25}{2} = 12\frac{1}{2}$ days.

Pipes and Cisterns.

1. **Inlet :-** A pipe connected with a tank or a cistern or a reservoir, that fills it, is known as an inlet.

2. **Outlet :-** A pipe connected with a tank or a cistern or a reservior, emptying it, is known as an outlet.

3. (i) If a pipe can fill a tank in x hours, then:
   part filled in 1 hour $= 1/x$.

   (ii) If a pipe can empty a full tank in y hours, then:
   part emptied in 1 hour $= 1/y$.

   (iii) If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (when $x > y$), then on opening both the pipes the net ~~the~~ part emptied in 1 hour

$$= \left(\frac{1}{y} - \frac{1}{x}\right).$$

Ex:-1 Two pipes A and B can fill a tank in 36 hours and 45 hours respectively. If both the pipes are opened simultaneously, how much time will be taken to fill the tank?

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Sol^n Part filled by A in 1 hour = $1/36$;

Part filled by B in 1 hour = $1/46$

Part filled by (A+B) in 1 hour = $\left(\frac{1}{36} + \frac{1}{46}\right) = \frac{9}{180}$

$= 1/20$.

Hence, both the pipes together will fill the tank in 20 hours.

Ex:-2 Two pipes can fill a tank in 10 hours and 12 hours respectively while a third pipe empties the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time will the tank be filled?

Sol^n Net part filled in 1 hour = $\left(\frac{1}{10} + \frac{1}{12} - \frac{1}{20}\right) = \frac{8}{60} = \frac{2}{15}$

$\therefore$ The tank will be full in $\frac{15}{2}$ hrs. = 7 hrs. 30 min

Ex:-3 If two pipes function simultaneously, the reservoir will be filled in 12 hours. One pipe fills the reservoir 10 hours faster than the other. How many hours does it take the second pipe to fill the reservoir?

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Sol^n Let the reservoir be filled by first pipe in x hours.
Then, second pipe will fill it in $(x+10)$ hours.

$$\therefore \frac{1}{x} + \frac{1}{x+10} = \frac{1}{12} \Rightarrow \frac{x+(x+10)}{x(x+10)} = \frac{1}{12}$$

$$\Rightarrow x^2 - 14x - 120 = 0 \Rightarrow (x-20)(x+6) = 0$$

$\Rightarrow x = 20$ (neglecting the -ve value of x)

So; the second pipe ~~at~~ will take $(20+10)$ hrs

i.e. 30 hrs to fill the reservoir.

Ex:-4 A cistern has two taps which fill it in 12 minutes and 15 minutes respectively. There is also a waste pipe in the cistern. When all the three are opened, the empty cistern is full in 20 minutes. How long will the waste pipe take to empty the full cistern?

Sol^n Work done by the waste pipe in 1 minute

$$= \frac{1}{20} - \left(\frac{1}{12} + \frac{1}{15}\right) = -\frac{1}{10}$$ (-ve sign means emptying)

$\therefore$ Waste pipe will empty the full cistern in 10 minutes.

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Ex:-5 An electric pump can fill a tank in 3 hours. Because of a leak in the tank it took $3\frac{1}{2}$ hours to fill the tank, If the tank is full, how much time will the leak take to empty it?

Sol^n Work done by the leak in 1 hour $= \left[\frac{1}{3} - \frac{1}{\left(\frac{7}{2}\right)}\right]$

$$= \frac{1}{3} - \frac{2}{7} = \frac{1}{21}.$$

The leak will empty the tank in 21 hours.

Ex:-6 Two pipes can fill a cistern in 14 hours and 16 hours respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom it took 32 minutes more to fill the cistern. When the cistern is full, in what time will the leak empty it?

Sol^n Work done by the two pipes in 1 hour $= \frac{1}{14} + \frac{1}{16} = \frac{15}{112}$

Time taken by these pipes to fill the tank $= \frac{112}{15}$ hrs

$= 7$ hrs. 28 min.

Due to leakage, time taken $= 7$ hrs. 28 min + 32 min.

$= 8$ hrs

~~Work done by the leak in 1 hour~~ $\cancel{= \left(\frac{15}{112} - \frac{1}{8}\right) = \frac{1}{112}}$

Work done by (two pipes + leak) in 1 hour $= 1/8$.
Work done by the leak in 1 hour $= \left(\frac{15}{112} - \frac{1}{8}\right) = \frac{1}{112}$.

$\therefore$ Leak will empty the full cistern in 112 hours.

Ex:-7 Two pipes A and B can fill a tank in 36 min. and 45 min. respectively. A water pipe C can empty the tank in 30 min. First A and B are opened. After 7 minutes, C is also opened. In how much time, the tank is full?

Sol^n Part filled in 7 min. $= 7\left(\frac{1}{36} + \frac{1}{45}\right) = \frac{7}{20}$

~~Remain~~ ~~Remain~~
Remaining part $= \left(1 - \frac{7}{20}\right) = \frac{13}{20}$.

Net part filled in 1 min. when A, B and C are opened $= \left(\frac{1}{36} + \frac{1}{45} - \frac{1}{30}\right) = \frac{1}{60}$.

Now $1/60$ part is filled in 1 min.

$\frac{13}{20}$ part is filled in $\left(60 \times \frac{13}{20}\right) = 39$ min.

Total time taken to fill the tank $= (39+7)$
$= 46$ min

Ex:- 8 Two pipes A and B can fill a tank in 24 min. and 32 min, respectively. If both the pipes are opened simultaneously, after how much time B should be closed so that the tank is full in 18 minutes?

Sol: Let B be closed after x ~~min~~ minutes. Then,

Part filled by (A + B) in x min. + Part filled by A in (18 - x) min. = 1.

$$\therefore x\left(\frac{1}{24}+\frac{1}{32}\right)+(18-x)\times\frac{1}{24}=1$$

$$\Rightarrow \frac{7x}{96}+\frac{18-x}{24}=1.$$

$$\Rightarrow 7x+4(18-x)=96$$

$$\Rightarrow x=8.$$

Hence, B must be closed after 8 minutes.